

scripts/scan-no-hardcoded-ipv4.sh

— User Guide

Last verified: 2026-05-15 (1.2.23 closure-cycle, §6.AD-debt task #61 backfill)
Inheritance: HelixConstitution §11.4.18 (script documentation mandate)

Overview

Stub doc generated from the script's in-source header comment. See `scripts/scan-no-hardcoded-ipv4.sh` for canonical behavior. This stub exists so the CM-SCRIPT-DOCS-SYNC pre-push gate (.github/hooks/pre-push Check 9) starts gating modifications to this script.

The script's own header documentation (verbatim):

```
scripts/scan-no-hardcoded-ipv4.sh – standalone §6.R IPv4 scanner.
```

Purpose: enforce the §6.R No-Hardcoding Mandate clause that no IPv4 literals appear in tracked source outside the exemption set.
§4.5.10 (CONTINUATION.md): the IPv4 enforcement is staged – this scanner is the mechanical gate that lands the rule.

Exit codes:

- 0 – no IPv4 violations
- 1 – IPv4 violation(s) found (paths printed to stderr)

Exemptions (kept in lockstep with the §6.R clause body):

<code>.env.example</code>	– placeholder file
<code>.lava-ci-evidence/</code>	– forensic anchors +
<code>matrix evidence</code>	
<code>docs/**/*md</code>	– design docs,
<code>plans, incident notes</code>	
<code>submodules/</code>	– submodules
<code>vendored at pinned hash</code>	
<code>*_test.go, *Test.kt, *Tests.kt, *Test.java</code>	– synthetic test
<code>fixtures</code>	
<code>src/test/, src/androidTest/</code>	– test source roots
<code>fixtures/</code>	– test HTML/JSON
<code>fixtures</code>	
<code>CHANGELOG.md</code>	– release notes may
<code>reference IPs in incident summaries</code>	
<code>*.md, *.json, *.xml, *.yaml, *.yml</code>	– external config +
<code>docs are legitimate</code>	home for

connection literals (Android
network_security_config.xml whitelists LAN
provisioning .yml lists service
Code that READS these files
targets.

ranges, Grafana
endpoints, etc.).
is what §6.R

Loopback / docs-prefix IPs that are universally permitted (RFC
5737,
documentation-only) are filtered AFTER the file-level exemption so
a
code file that legitimately uses 127.0.0.1 or 0.0.0.0 still passes
—
those are not "hardcoded connection addresses" in the §6.R sense
(they cannot connect to anything but the local host or the
wildcard).

Usage

See the script's in-source comment block (above) for canonical usage examples.

Maintenance

When this script is modified, update this document in the same commit (CM-SCRIPT-DOCS-SYNC requires it). Per §11.4.18, the documentation **MUST** stay in sync with the codebase — no doc may be out of sync with its script.

Cross-references

- scripts/scan-no-hardcoded-ipv4.sh — the script itself
- docs/helix-constitution-gates.md — gate inventory
- HelixConstitution Constitution.md §11.4.18 (the mandate)
- Lava CLAUDE.md §6.AD (HelixConstitution Inheritance)